Session 1: Foundations of Marine Macroecology and Niche Theory

Modeling the distribution of biodiversity under global climate change

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Introduction to Species Distribution Modeling

What is species distribution modeling?

Also known as Environmental Niche Modelling, Habitat Suitability Modelling, …

Process of **using computer algorithms to estimate species ecological niches and predict the distributions** (occurrence, abundance) **across space and time**.

Purpose:

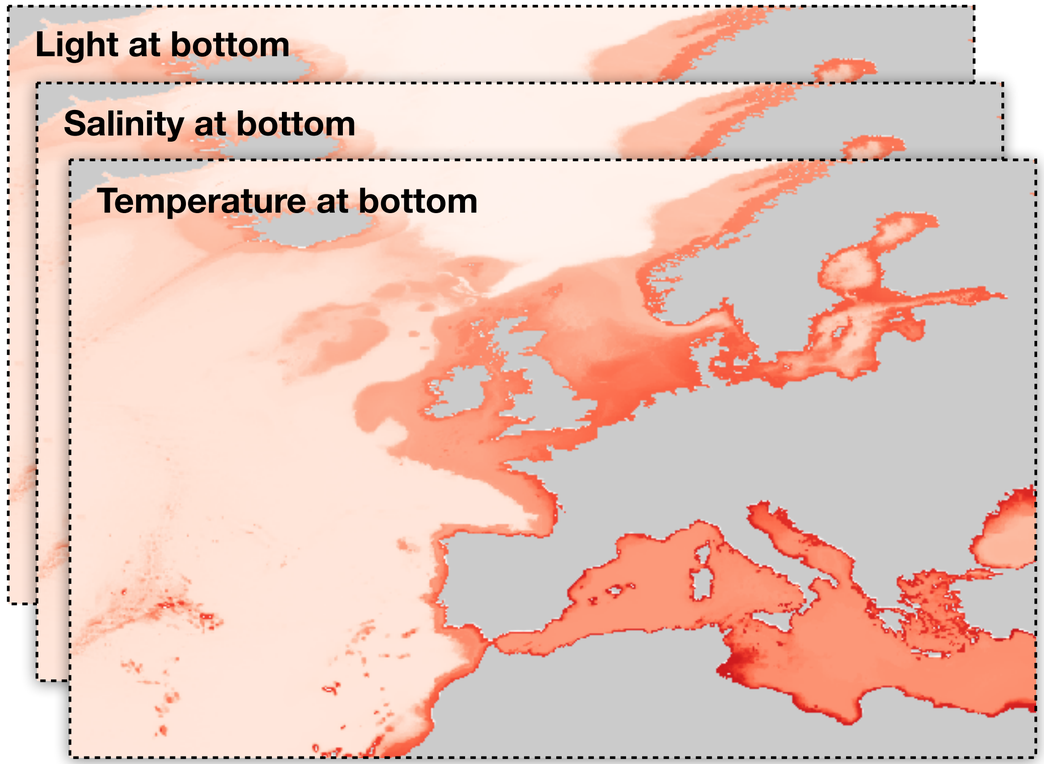
Gain insights into species' environmental tolerances and habitat preferences.

Make spatial predictions of geographic distributions.

Main SDM approaches

Mechanistic modeling: uses **information about the physiological response of species** to environmental conditions (e.g., temperature limits for survival). Require empirical data (often from lab experiments), which may not always be available.

Correlative modeling: based on the **statistical correlation between species occurrence records** (presence-absence) **and the environment**.



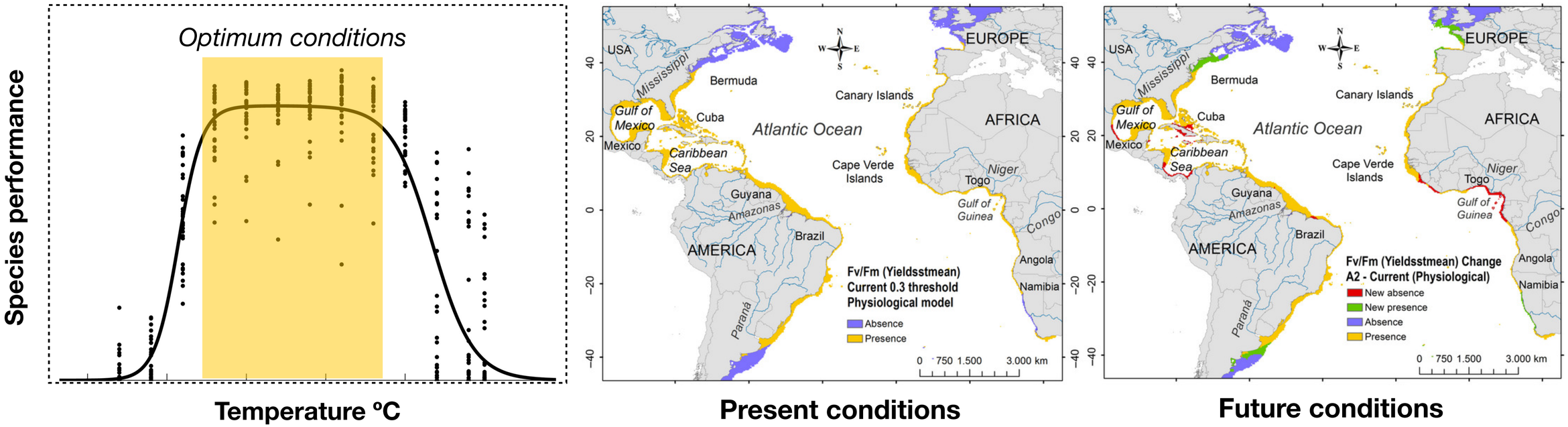


Mechanistic distribution models

Process based: built by reclassifying environmental gradients with tolerance limits.

A straightforward approach to predict the distribution of species.

Presence = [Light > 5 $E.m^2.year^{-1}$] ∩ [5 <= Temp. (ºC) <= 20.5ºC] ∩ [Sal. > 28]

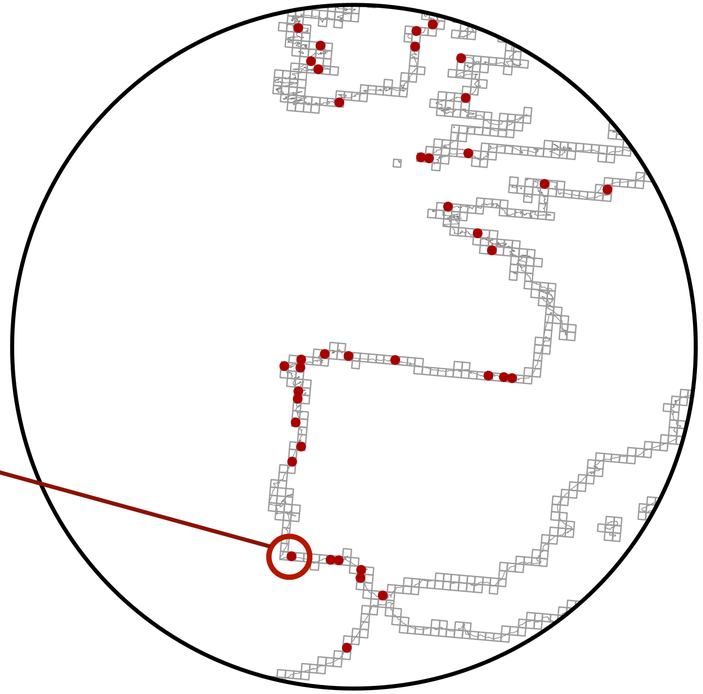


Mechanistic distribution models

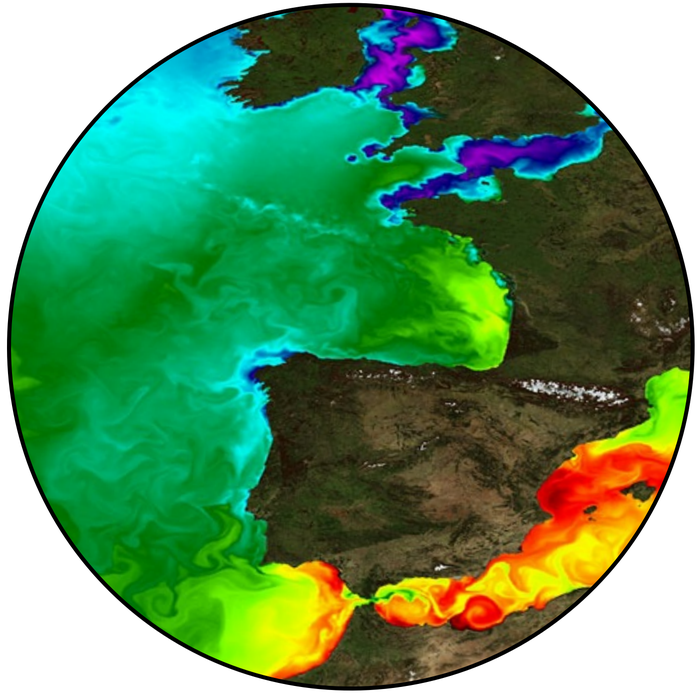
Data driven mechanistic distribution models: (more realistic) by **using response curves from experiments** determining the tolerance of species - from cold and warm conditions (e.g., a species' photosynthetic performance measured by maximal quantum yield) - the optimal conditions can be mapped.

Rodríguez et al., 2019

Obs.	SST	Salinity	Oxygen
1	21.0	35.4	210.1
1	19.4	29.1	224.9
1	21.5	31.1	199.8
0	21.3	33.4	201.1
0	14.8	34.1	184.8
(…)	(…)	(…)	(…)



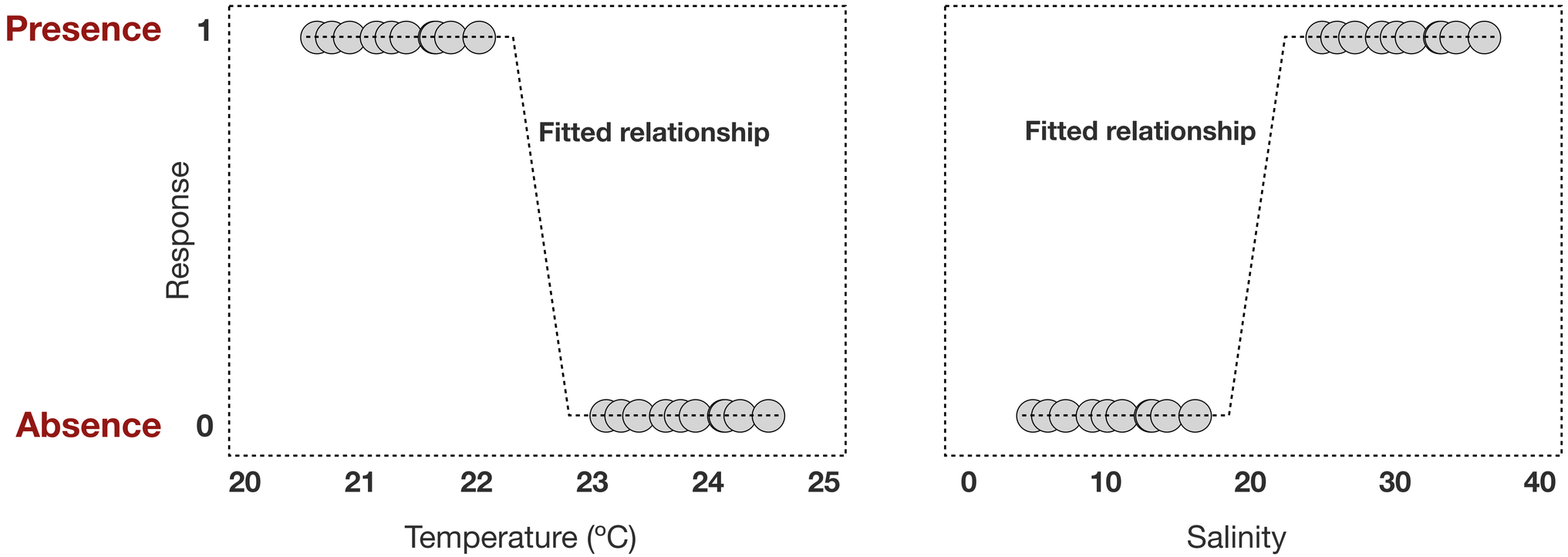
Records of species distribution



Environmental data

Correlative distribution models

Use **statistical correlations between observed species occurrence records and the environmental conditions** at those locations.



The **modeling algorithm learns the effect of environmental** variables **on the distribution of species** (**training data**) and **fits a relationship**.

Connecting niche theory to SDMs

Correlative SDMs typically model an 'A'-based niche (of the BAM diagram), thus capturing the abiotic drivers of distributions, and assuming or ignoring B and M.

Under the core assumptions: (1) the observed **geographic distribution is an indicator of a species' environmental requirements**; (2) **niche conservatism** (e.g., to be able to predict future distributions under climate change).

Terrestrial vs. Marine Niche Modeling

Same technical approaches and assumptions. Minor detail changes.

Dimensionality: Terrestrial models often utilize 2D surfaces (precipitation, air temp). Marine models must account for a 3D environment (surface vs. benthic depths).

Dispersal: Terrestrial mostly relies on dispersal across static landscapes; marine relies heavily on passive transport via fluid, dynamic ocean currents.

Detectability & bias: Marine environments (especially the deep sea) suffer from severe sampling biases and inaccessibility, leading to poor occurrence data compared to the terrestrial realm.